

RESEARCH NOTE

Analysis of Gas Behaviors Under Ideal and Real Conditions

C. Zeng*

North Carolina School of Science and Math, Durham, NC

*Corresponding author. Email: zeng25c@ncssm.edu

Abstract

The applications of computational science as a set of technologies, techniques, and tools are significant in helping scientists understand natural processes. The section on computational chemistry allows for the analysis of gas behaviors, learning about its properties and states under dynamic conditions, and having real-world applications such as the pressing issue of climate change. In the high school curriculum, chemistry courses teach the ideal gas law to model the relationship between pressure, temperature, and the volume of gases. However, the ideal gas law is used for simplistic purposes using a hand-held calculator and fails to implement important elements of gas molecules. In this study, real gas laws including Van der Waals and Redlich-Kwong are compared to the ideal gas law to show its inaccuracies and present real gas laws as improved models that accommodate for additional factors.

Keywords: Gas laws; Van Der Waals; Redlich-Kwong; Ideal Gas Law; Real Gas Laws; Mathematica

1. Introduction

Collecting and analyzing information on gas behaviors is crucial in the study of chemistry, biology, physics, and other sciences, providing important insights into the natural world. From the surrounding air to chemical experiments to urgent issues, gases play a monumental role in biological processes and global policy. Specifically, recent anthropogenic activities in the past decades have caused a drastic rise in the greenhouse effect, leading to global warming, rising sea levels, and climate change, all originating from gases trapped in the troposphere. Gases are composed of individual atoms and molecules, which display different behaviors based on the dynamic relationships of different variables, including pressure, temperature, and volume. We need to study them to gain a better comprehension of the physical properties that make up the surroundings around us. Therefore fundamental principles known as gas laws have been constructed to describe these interactions, understanding and predicting phenomena.

In high school chemistry classrooms or labs, gases are studied throughout the curriculum as they serve as foundational principles for other, more complex concepts. Teachers educate students on the states, properties, and behaviors of gases to grasp ideas in chemical reactions, results from chemical experiments, and possible phenomena that occur in their daily lives. They must comprehend the building blocks of environmental processes, which are becoming more relevant every day with disturbing problems such as climate change. During the learning process, students develop critical thinking skills and potential interests that motivate them to further pursue research. Therefore, the

study of gases in high school provides the framework for future undergraduate research in adjacent fields such as chemistry, meteorology, environmental science, etc. Learning about gases at the high school level is important because of its real-world applications and helping students be more informed of chemical principles.

Gases are taught in the classroom according to the ideal gas law. Over the course of scientific history, laws have been established in an attempt to represent variable conditions and have led to many interpretations of gas characteristics and concepts. Early historical discoveries include Boyle’s law in 1662, which proposed through experiments that volume was inversely proportional to the pressure at a constant temperature. Similarly, Charles’ and Gay-Lussac’s law in the late 18th century stated that temperature was proportional to volume at standard pressure, and Avogadro’s principle in the early 19th century claims volume to be proportional to the number of molecules (Laugier and Garai 2007). From early principles, the ideal gas law emerged as a combination of these foundational theories to incorporate pressure, temperature, and volume into a comprehensive equation.

Although the ideal gas law effectively measures relationships in ideal conditions, it was founded upon unrealistic assumptions, causing it to be less reliable in real-world applications. The two assumptions are that measured gas molecules never collide and that they are infinitely compressible, or as the pressure increases, the volume approaches zero. When the compressibility factor Z , or $\frac{PV}{RT}$, is at lower pressures or temperatures, molecules are more drawn by attractive forces, causing Z to be less than ideal behavior; at higher temperatures, as molecules become primarily influenced by their repulsive forces, $\bar{V}$, or the molar volume, is greater than $\bar{V}$ as calculated in the ideal gas law (McQuarrie and Simon 1997). The reason behind these simplifying approaches is for high school chemistry students to understand the concept more easily, transforming complex cubic functions into straightforward algebraic relationships.

In this study, we present how real gas laws such as Van der Waals and Redlich-Kwong can be used to analyze the differences between gas laws in ideal and real conditions. This research paper calculates data tables and graphs to visualize discrepancies of pressure in atmospheres (atm) based on the changing molar volume (L/mol) at standard temperature and other various constants. The purpose of the research paper is to emphasize the inaccuracies of the ideal gas law compared to more real-world applicable equations that account for other variable factors affecting the behaviors of gases. Specifically, the behavior of methane molecules (CH_4) is measured to reach conclusions about the differences between ideal and real gas laws.

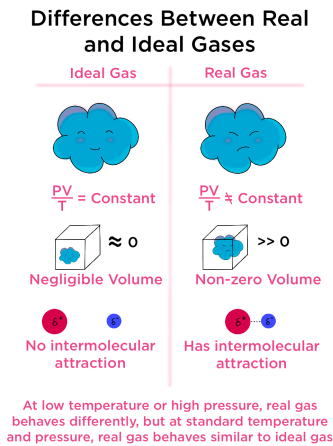


Figure 1. Chart Comparing Real and Ideal Gas Laws (expii 2024)

2. Computational Approach

Equation 1 shows the ideal gas law (McQuarrie and Simon 1997):

$$P = \frac{RT}{\bar{V}} \quad (1)$$

where:

- P is pressure in atmospheres,
- T is temperature in K,
- $\bar{V}$ is the molar volume in liters per mole,
- R is the universal gas constant, which is defined at $8.314 \frac{J}{mol \cdot K}$ when pressure is in Pascals and volume is in cubic meters, and $0.0821 \frac{L \cdot atm}{mol \cdot K}$ when pressure is in atmospheres and volume is in liters

Expanding, new innovative approaches to describe gas behaviors present their own advantageous features. Van der Waals algorithm, proposed by Van der Waals, the grandfather of the non-ideal gas laws, in 1873 within his doctoral dissertation, added new collision and compressibility constants that better take into account the complex dynamics of the environment, denoted by a and b respectively (Cropper 2012). Additionally, when constants a and b are determined for a substance, Van Der Waals can be used to predict boiling points, critical points, and other attributes.

Equation 2 shows the Van der Waals gas law (McQuarrie and Simon 1997):

$$P = \frac{RT}{\bar{V} - b} - \frac{a}{\bar{V}^2} \quad (2)$$

where:

- P is pressure in atmospheres,
- T is temperature in K,
- $\bar{V}$ is the molar volume in liters per mole,
- R is the universal gas constant,
- a is the constant accounting for attractive forces between molecules,
- b is the constant accounting for the effective volume of the molecules themselves

Similarly, other scientists have developed more accurate equations that define gas behaviors, including Redlich-Kwong, Peng-Robinson, and Beattie-Bridgeman gas laws (McQuarrie and Simon 1997). Equation 3 shows the Redlich-Kwong gas law (McQuarrie and Simon 1997):

$$P = \frac{RT}{\bar{V} - b} - \frac{a}{\sqrt{T}\bar{V}(\bar{V} + b)} \quad (3)$$

where:

- P is pressure in atmospheres,
- T is temperature in K,
- $\bar{V}$ is the molar volume in liters per mole,
- R is the universal gas constant,
- a is the constant accounting for attractive forces between molecules,
- b is the constant accounting for the effective volume of the molecules themselves

The Redlich-Kwong gas law introduces the term $\frac{a}{\sqrt{T}}$, which incorporates temperature dependency, allowing for more accuracy at higher temperatures. Additionally, $\bar{V} + b$ applies to a greater range of volumes for behaviors of different gases (OpenAI 2024).

We used Mathematica to conduct this study. Mathematica is a software program that allows for computation, data analysis, and implementation of programming languages. It is owned by the company of Wolfram Research, founded by Stephen Wolfram in 1987 focuses on computational sciences, and offers various services including programming languages, engines, data storage, and more (OpenAI 2024). The Wolfram Language is the underlying programming language within the Mathematica software. Divided into the kernel and the front end, interpreted functions in the kernel that are displayed in the front end provide for high-performance computation and the ability for the interface to add extensions and other applications (Wikipedia contributors 2024b).

In Mathematica, we established two variable gas law functions depending on temperature and molar volume, created a results table, and plotted the different equations with different features. The results tables displays values at the changing temperatures, with intervals of volume of 0.1 L from 0 L to 2 L. Built-in programming commands allowed for easy modifications and better visualization of the plot, including grid-lines, color styles, labels, and a legend.

Sample Mathematica commands included:

1. Grid: This command was used to format the data table into a neat visualization.
2. Grid-lines: This command was used to create grid-lines on the graph.
3. Plot: This command was used to plot the functions.
4. PlotLabel/AxisLabel: These commands were used to label the graph with appropriate variables.
5. PlotLegends: This command created a legend for the graph that listed the different gas laws plotted.
6. PlotStyle: This command was used to color code the different equations.
7. Table: This command was used to create a results table.

Possible experiments include:

1. Compare each of the five gases at a constant temperature and volume
2. Compare ONE gas at a constant temperature with a changing volume
3. Compare ONE gas at a constant volume with a changing temperature

3. Results

3.1 Experiment 1: Temperature at 273.15 K

Figure 2 shows the results for an analysis of CH_4 going from volume of 0 L to 2 L at a temperature of 273.15 K, or 0 C, the freezing point of water.

Volume (L)	Ideal Gas Pressure (atm)	Van Der Waals Pressure (atm)	Redlich-Kwong Pressure (atm)
0.01	2241.47	-23 392.	-5955.12
0.11	203.77	149.796	154.647
0.21	106.737	83.1787	86.2412
0.31	72.3054	60.4942	61.7556
0.41	54.67	47.6579	48.2988
0.51	43.9504	39.3226	39.6977
0.61	36.7454	33.4672	33.7088
0.71	31.57	29.1278	29.2945
0.81	27.6725	25.7833	25.9043
0.91	24.6315	23.127	23.2183
1.01	22.1928	20.9665	21.0375
1.11	20.1934	19.1748	19.2315
1.21	18.5245	17.665	17.7112
1.31	17.1104	16.3754	16.4138
1.41	15.8969	15.2612	15.2935
1.51	14.8442	14.2889	14.3164
1.61	13.9222	13.433	13.4568
1.71	13.108	12.6739	12.6945
1.81	12.3838	11.9959	12.014
1.91	11.7354	11.3867	11.4027

Figure 2. Results table for CH_4 with volume going from 0 to 2 L at 273.15 K

Figure 3 shows a graph comparing the different gas laws analyzing the behavior of CH_4 going from volume of 0 L to 2 L at a temperature of 273.15 K.

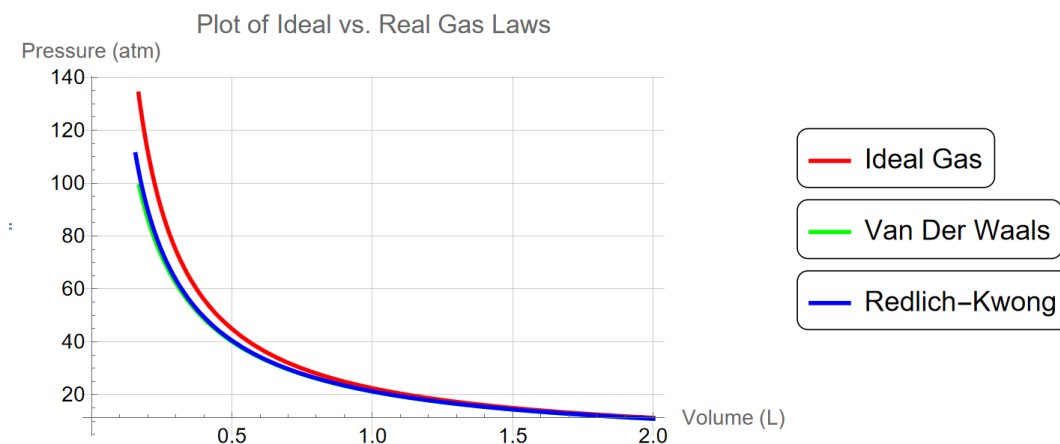


Figure 3. Graph for CH_4 with volume going from 0 to 2 L at 273.15 K

Analysis: At this temperature, we see that there are slight differences in the gas pressure as the volume changes. During lower volumes less than 1 L, both Van der Waals and Redlich-Kwong equations predict lower pressure of CH_4 at the same temperature and volume than what the ideal gas law calculated. However, there are key similarities between the real and ideal equations, as when the volume becomes astronomically large, $\bar{V} - b$ is indifferent from $\bar{V}$ and $\frac{a}{\bar{V}^2}$ approaches zero, making

it the same form as the ideal gas law (Wikipedia contributors 2024a). Therefore, from the graph, as the volume increases, all three gas laws converge and approach similar values.

3.2 Experiment 2: Temperature at 298.15 K

Figure 4 shows the results for an analysis of CH_4 going from volume of 0 L to 2 L at a standard temperature of 298.15 K.

Volume (L)	Ideal Gas Pressure (atm)	Van Der Waals Pressure (atm)	Redlich–Kwong Pressure (atm)
0.01	2446.62	−23453.	−5851.71
0.11	222.42	180.696	185.599
0.21	116.506	95.508	99.2647
0.31	78.9232	68.1952	69.8605
0.41	59.6736	53.2571	54.1522
0.51	47.9729	43.7213	44.2696
0.61	40.1085	37.0893	37.4561
0.71	34.4594	32.2063	32.4676
0.81	30.2052	28.4602	28.655
0.91	26.8859	25.4949	25.6455
1.01	24.2239	23.0893	23.209
1.11	22.0416	21.0985	21.1959
1.21	20.22	19.4238	19.5044
1.31	18.6765	17.9954	18.0632
1.41	17.3519	16.7626	16.8205
1.51	16.2028	15.6879	15.7379
1.61	15.1964	14.7427	14.7863
1.71	14.3077	13.905	13.9432
1.81	13.5172	13.1573	13.1911
1.91	12.8095	12.4859	12.5161

Figure 4. Results table for CH_4 with volume going from 0 to 2 L at 298.15 K

Figure 5 shows a graph comparing the different gas laws analyzing the behavior of CH_4 going from volume of 0 L to 2 L at a standard temperature of 298.15 K.

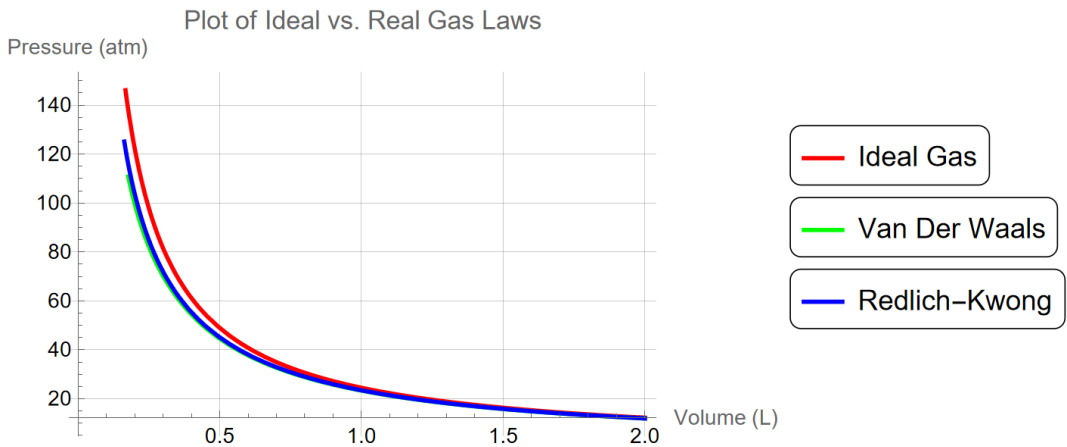


Figure 5. Graph for CH_4 with volume going from 0 to 2 L at 298.15 K

Analysis: At this temperature, we see that the predicted pressure has many of the same characteristics as from Experiment 1 at 273.15 K. Both experiments show Van der Waals and Redlich-Kwong equations having similar graphical trends, predicting lower pressure of CH_4 at the same temperature and volume compared to the ideal gas law, and all three gas laws converging after approximately 1 L.

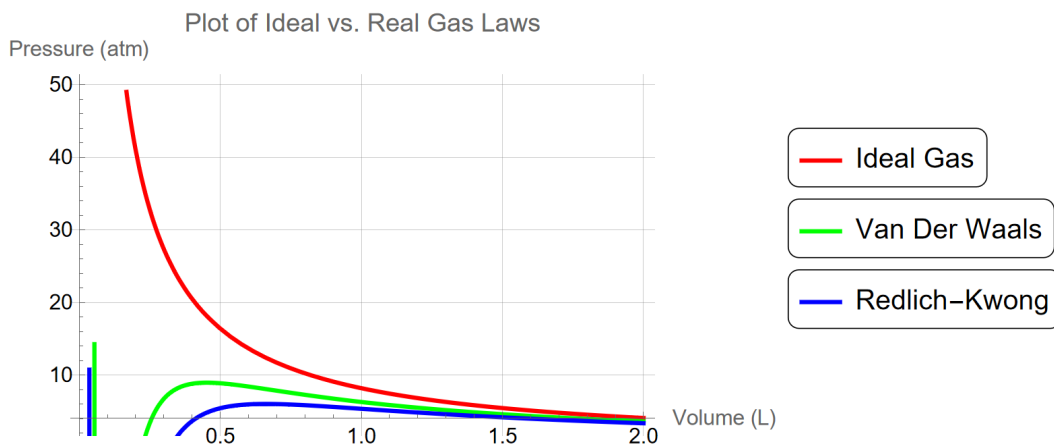
3.3 Experiment 3: Temperature at 100 K

Figure 6 shows the results for an analysis of CH_4 going from volume of 0 L to 2 L at a temperature of 100 K.

Volume (L)	Ideal Gas Pressure (atm)	Van Der Waals Pressure (atm)	Redlich-Kwong Pressure (atm)
0.01	820.6	-22969.2	-8389.31
0.11	74.6	-64.2126	-104.228
0.21	39.0762	-2.21363	-17.552
0.31	26.471	7.15687	-0.877457
0.41	20.0146	8.87798	3.96158
0.51	16.0902	8.85758	5.54625
0.61	13.4525	8.38094	6.00132
0.71	11.5577	7.80602	6.01427
0.81	10.1309	7.24365	5.84629
0.91	9.01758	6.72722	5.60713
1.01	8.12475	6.26365	5.34586
1.11	7.39279	5.85069	5.08499
1.21	6.78182	5.48322	4.83473
1.31	6.26412	5.1556	4.59934
1.41	5.81986	4.86254	4.38016
1.51	5.43444	4.59938	4.17708
1.61	5.09689	4.36209	3.98931
1.71	4.79883	4.14725	3.81577
1.81	4.5337	3.95197	3.65529
1.91	4.29634	3.77379	3.5067

Figure 6. Results table for CH_4 with volume going from 0 to 2 L at 100 K

Figure 7 shows a graph comparing the different gas laws analyzing the behavior of CH_4 going from volume of 0 L to 2 L at a standard temperature of 100 K.



Analysis: At the temperature of 100 K, there are clear differences between the real gas laws and the ideal gas law. The graph shows a clear representation of the complexity of the real gas laws and its equation state as cubic functions dependent on the molar volume (McQuarrie and Simon 1997), compared to the simple relationship given by the ideal gas equation. As the pressure of CH_4 calculated in the ideal gas law consistently decreases, both the Van der Waals and Redlich-Kwong gas laws predict a rapid decrease followed by an increase that slowly plateaus out and converges with ideal gas pressure. The analysis shows that the ideal gas law is more inaccurate at colder temperatures and its assumptions cause a simplistic depiction of gas behaviors.

3.4 Experiment 4: Temperature at 373.15 K

Figure 8 shows the results for an analysis of CH_4 going from volume of 0 L to 2 L at a temperature of 373.15 K, or 100 C.

Volume (L)	Ideal Gas Pressure (atm)	Van Der Waals Pressure (atm)	Redlich-Kwong Pressure (atm)
0.01	3062.07	-23 636.1	-5671.54
0.11	278.37	273.394	275.085
0.21	145.813	132.496	137.306
0.31	98.7764	91.2983	93.6833
0.41	74.6846	70.0546	71.4251
0.51	60.0406	56.9172	57.797
0.61	50.1979	47.9554	48.565
0.71	43.1277	41.4419	41.8882
0.81	37.8033	36.4906	36.8311
0.91	33.6491	32.5985	32.8665
1.01	30.3175	29.4578	29.6742
1.11	27.5862	26.8699	27.0481
1.21	25.3064	24.7003	24.8497
1.31	23.3746	22.8552	22.9822
1.41	21.7168	21.2668	21.376
1.51	20.2786	19.885	19.9799
1.61	19.0191	18.6718	18.7551
1.71	17.9068	17.5983	17.6719
1.81	16.9175	16.6415	16.7071
1.91	16.0318	15.7834	15.8422

Figure 8. Results table for CH_4 with volume going from 0 to 2 L at 373.15 K

Figure 9 shows a graph comparing the different gas laws analyzing the behavior of CH_4 going from volume of 0 L to 2 L at a temperature of 373.15 K.

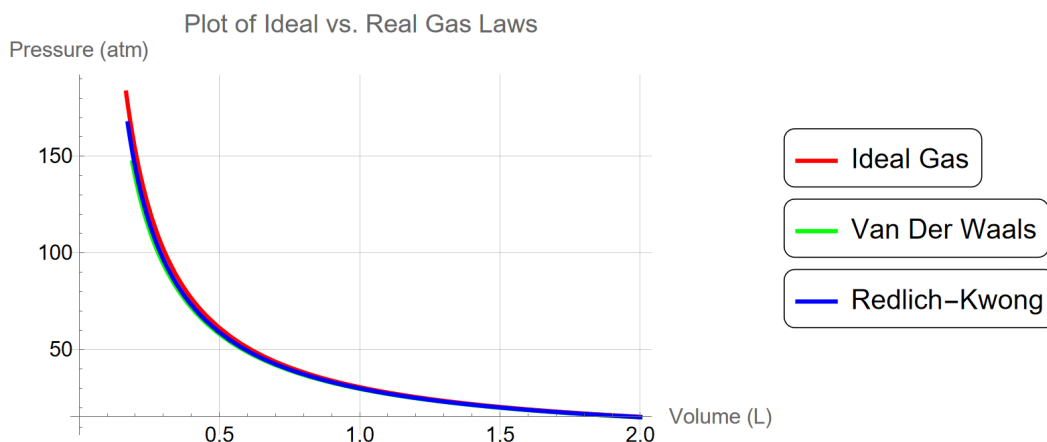


Figure 9. Graph for CH_4 with volume going from 0 to 2 L at 373.15 K

Analysis: At this temperature, we see that the pressure of CH_4 behaves homogeneously across all three gas laws, exponentially decaying and approaching zero. The data table and plot show that the ideal gas law can be utilized at hotter temperatures as it has analogous characteristics that are still present in the Van der Waals and Redlich-Kwong gas laws.

4. Discussion

Real gas laws including Van der Waals and Redlich-Kwong present a better representation of gas behaviors under different conditions than the ideal gas law taught in high school chemistry classrooms. From the result tables and graphs, most divergences between the ideal and real gas equations are found at lower volumes and colder temperatures. Although there are some instances where the ideal gas law is an effective model that yields similar results, the overall analysis of the ideal gas law concludes that it fails to account for the collision and volume of molecules, causing discrepancies in the calculated pressure with changing temperatures and volumes.

Understanding gas behaviors has increasing relevancy in real-world applications, in and out of the classroom. Educating high school students with higher information provides them with an accurate knowledge base, personal growth, and professional preparation for future careers. Additionally, the increasing problem of climate change requires improved scientific research on gas interactions and relationships that correctly simulate the environment. We anticipate that future progress in computational chemistry will lead to improvements in current gas models that expand the general foundation of gases in the natural world.

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Appendix 1. Source Code

```

idealPlot =
Plot[igl[vbar, temp], {vbar, minVolume, maxVolume},
PlotLabel -> "Plot of Ideal Gas Law",
AxesLabel -> {"Volume (L)", "Pressure (atm)"},
GridLines -> Automatic, PlotStyle -> Red,
PlotLegends -> LineLegend[{"Ideal Gas"}, LegendFunction -> Frame]]

vdwPlot =
Plot[vdw[vbar, temp], {vbar, minVolume, maxVolume},
PlotLabel -> "Plot of Van Der Waals Gas Law",
AxesLabel -> {"Volume (L)", "Pressure (atm)"},
GridLines -> Automatic, PlotStyle -> Green,
PlotLegends ->
LineLegend[{"Van Der Waals"}, LegendFunction -> Frame]]

rkPlot =
Plot[rk[vbar, temp], {vbar, minVolume, maxVolume},
PlotLabel -> "Plot of Redlich-Kwong Gas Law",
AxesLabel -> {"Volume (L)", "Pressure (atm)"},
GridLines -> Automatic, PlotStyle -> Blue,
PlotLegends ->
LineLegend[{"Redlich-Kwong"}, LegendFunction -> Frame]]

Show[idealPlot, vdwPlot, rkPlot,
PlotLabel -> "Plot of Ideal vs. Real Gas Laws"]

```